

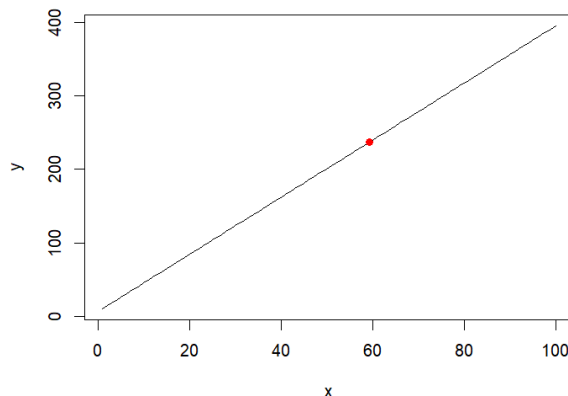
- 3.18** A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = 6.855 + 3.880INCOME$$

(se) (7.383) (0.112)

- a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of *INCOME* = 59.3. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.
- c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .
- d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.
- e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

a.



Sample mean of amount of insurance held is

$$6.855 + 3.880 * 59.3 = 236.939$$

b.

Income increase \$1000, insurance will increase \$3880 in average

Point estimates: $3.880 * 1000 = 3880$

$df = 20 - 2 = 18$, $t_c = 2.1$

95% int : $[3.880 - 2.1 * 0.112, 3.880 + 2.1 * 0.112] = [3.6448, 4.1152] =$

[\$3644.8, \$4115.2]

c. 99% CI, $100 \cdot \text{INCOME}$, $\text{cov} = -0.746$, $t_c = 2.878$

$$L = 6.855 + 100 \cdot 3.880 = 394.855$$

$$\text{var}(L) = \text{var}(b_1) + 100^2 \text{var}(b_2) + 2 \cdot 100 \cdot \text{cov}(b_1, b_2)$$

$$= 7.383^2 + 100^2 \cdot 0.112^2 + 2 \cdot 100 \cdot -0.746 = 30.74869$$

$$\text{se}(L) = \sqrt{30.74869} = 5.545$$

$$\text{CI}(99\%) = [394.855 - 2.878 \cdot 5.545, \quad 394.855 + 2.878 \cdot 5.545]$$

$$= [378.8965, 410.8135] = [\$378896.5, \$410813.5]$$

d.

$$H_0: \beta_2 = 5, \quad H_A: \beta_2 \neq 5$$

$$t = \frac{3.88 - 5}{0.112} = -10$$

$$t_{97.5\%, 18} = \pm 2.1 (\text{reject region is any } t > 2.1 \text{ or } t < -2.1)$$

Because $t(-10) < t_{97.5\%, 18} (= -2.1)$, therefore t falls in the rejection region, we reject H_0

The data not support the claim.

e.

$$H_0: \beta_2 = 1, \quad H_A: \beta_2 > 1$$

$$t = \frac{3.88 - 1}{0.112} = 25.71$$

$$t_{99\%, 18} = 2.55 (\text{rejection region is any } t > 2.55)$$

Because $t(= 25.71) > t_{99\%, 18} = 2.55$, therefore t falls in the rejection region, we reject H_0

The data not support the claim, income increase and insurance increase is not the same amount.

3.23 The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

- a. Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p -value. What do you conclude?
- b. Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000.

Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p -value. What do you conclude?

- c. Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.
- d. Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

a.

$$y = \alpha_1 + \alpha_2 x^2 + e$$

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	93.565854	6.072226	15.41	<2e-16 ***
I(sqft^2)	0.184519	0.005256	35.11	<2e-16 ***

$$\frac{dy}{dx} = 2\alpha_2 x = 40\alpha_2$$

$$\text{claims that } \frac{dy}{dx} = 13$$

$$\alpha_2 = \frac{13}{2 * 20} = 0.325$$

$$H_0: \alpha_2 \leq 0.325, \quad H_A: \alpha_2 > 0.325$$

$$df=500-2=498$$

$$t = \frac{0.184519 - 0.325}{0.005256} = -26.73$$

$$t_{95\%,498} = 1.65 (\text{rejection region is } t > 1.65)$$

$$p \text{ value} = 1$$

We do not reject H_0 because $p\text{-value}=1 > 5\%$, the marginal effect is less than \$13000.

b.

$$sqft = x = 40$$

$$\alpha_2 = \frac{13}{2 * 40} = 0.1625$$

$$H_0: \alpha_2 \leq 0.1625, \quad H_A: \alpha_2 > 0.1625$$

$$df=500-2=498$$

$$t = \frac{0.184519 - 0.1625}{0.005256} = 4.1893$$

$$t_{95\%,498} = 1.65 (\text{rejection region is } t > 1.65)$$

$$p \text{ value} = 1.655124e^{-05}$$

We reject H_0 because $p\text{-value} < 5\%$, the marginal effect is larger than \$13000.

c.

point est: $L = E(\text{price}|\text{sqft} = 20) = 93.565854 + 0.184519 * 20^2 = 167.3735$

```
> vcov(mod)
              (Intercept)      I(sqft^2)
(Intercept) 36.87192362 -2.345446e-02
I(sqft^2)    -0.02345446  2.762344e-05
```

$$\text{var}(L) = \text{var}(a_1) + (20^2)^2 \text{var}(a_2) + 2 * 20^2 \text{cov}(a_1, a_2) = 22.528$$

$$\text{se}(L) = 4.75$$

$$t_{97.5\%, 498} = 1.965$$

$$\text{CI}(95\%) = [167.3735 - 1.965 * 4.75, 167.3735 + 1.965 * 4.75] = [158.0397, 176.7073] = [\$158039.7, \$176707.3]$$

Repeated sampling, about 95% interval will cover true average price with 2000 sqft

d.

The data: 138 169 183

$$\text{mean} = 163.333$$

Yes, compatible with (c) because 163.333 fell in the CI(95%)

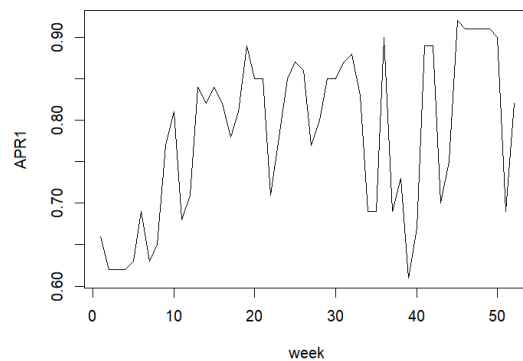
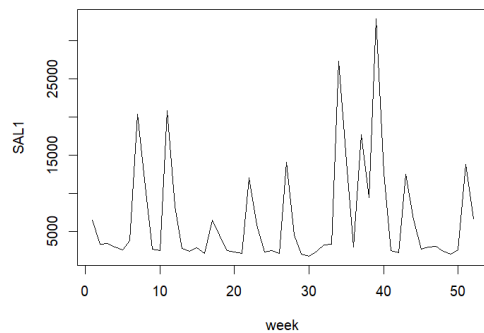
3.31 Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file *tuna*. There are 52 observations for each of the variables. The variable *SAL1* = unit sales of brand no. 1 canned tuna, and *APR1* = price per can of brand no. 1 tuna (in dollars).

- Calculate the summary statistics for *SAL1* and *APR1*. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus *WEEK*. How much variation in sales and price is there from week to week?
- Plot the variable *SAL1* (y-axis) against *APR1* (x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?
- Create the variable $\text{PRICE1} = 100\text{APR1}$. Estimate the linear regression $\text{SAL1} = \beta_1 + \beta_2 \text{PRICE1} + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?
- Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.
- Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 “at the means.” Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?
- Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the *p*-value for the test. What is your conclusion?

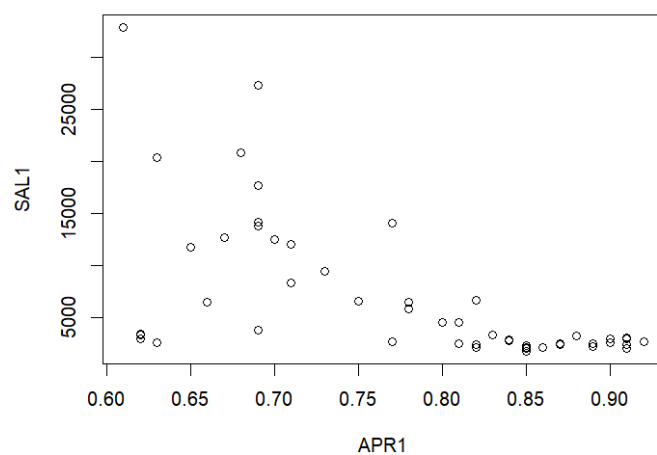
a.

	Mean	Min	Max	std
SAL1	6718.71	1762	32820	6915.08

APR1	0.78	0.61	0.92	0.1
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The time series plot shows that SAL1 and APR1 fluctuates considerably across weeks
b.



There is an inverse relationship. As expected, the higher the price, the lower the sales.

c.

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	40714.22	6196.42	6.571	2.82e-08	***
price1	-434.45	78.58	-5.528	1.17e-06	***

point estimate : when the 1 cent price increase, the sale will decrease 434.45 in average.

df = 52-2=50, $t_{97.5\%,50} = 2.00856$

CI(95%) : $[-434.45 - 2.00856 * 78.58, -434.45 + 2.00856 * 78.58] = [-592.3, -276.6]$

d. price1=70

$$L = E[SAL1|PRICE1 = 70] = 40714.22 - 434.45 * 70 = 10302.72$$

```
> vcov(mod)
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```
              (Intercept)      price1
(Intercept) 38395634.2 -483239.567
price1      -483239.6   6175.586
```

$$\text{var}(L) = 38395634 + 70^2 * 6175 + 2 * 70 * -483239 = 999674$$

$$\text{se}(L) = 999.837$$

$$t_{95\%,50} = 1.676$$

$$\text{CI}(90\%) : [10302.72 - 1.676 * 999.837, 10302.72 + 1.676 * 999.837] = [8627, 11978.45]$$

e.

$$\text{elasticity} = b_2 \frac{\text{mean}(\text{price1})}{\text{mean}(\text{SAL1})} = -434.45 \frac{78.25}{6718.712} = -5.06$$

$$\text{df} = 52 - 2 = 50, t_{97.5\%,50} = 2.00856$$

$$\text{CI}(95\%) : \frac{78.25}{6718.712} * [-592.3, -276.6] = [-6.9, -3.22]$$

The sales decrease about 3.22% to 6.9% when price increase by 1 %, they are fairly elastic.

Yes, it makes economic sense. Because canned tuna is not a necessity for a normal family, when its price increases, people tend to buy other foods.

f.

$$H_0: e = -3 \quad H_A: e \neq -3$$

$$\beta_{2,0} = -3 \frac{6718.712}{78.25} = -257.6$$

Equivalently

$$H_0: \beta_2 = -257.6, \quad H_A: \beta_2 \neq -257.6$$

$$t_{95\%,50} = 1.676, \text{ rejection region is any } |t| > 1.676$$

$$t = \frac{-434.45 + 257.6}{78.58} = -2.25$$

$$\text{p-value: } 0.014$$

we reject H_0 because $\text{p-value} = 0.014 < 5\%$